Annexure A - Concept Mapping

Cryptography and Network Security | Assessment Tool 2 | CO1
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Q1. Security Attacks, Threats, Risks and Security Goals

Concept map flow: Security Attacks (Passive: eavesdropping, traffic analysis | Active: masquerade, replay, modification, DoS/DDoS) exploit vulnerabilities, creating Threats (interception, modification, fabrication, interruption), which lead to Risks. Security Goals (Confidentiality, Integrity, Availability) mitigate them.

(a) Attack to Threat to Risk

Attack	Type	Threat created	Risk	Real-world example
Eavesdropping	Passive	Interception	Loss of confidential data, identity theft	Sniffing unencrypted public Wi-Fi traffic
Traffic analysis	Passive	Leakage of communication patterns	Privacy loss	Observing who talks to whom and when
Phishing / masquerade	Active	Impersonation	Account takeover, financial loss	Fake bank login page
Replay attack	Active	Fabrication	Duplicate transactions	Re-sending a captured payment request
Modification	Active	Modification	Data corruption, fraud	Altering the amount in a transfer
DoS / DDoS	Active	Interruption	Service downtime	Botnet flooding a website

(b) Security goal mitigates attack/threat

Security goal	What it means	Mechanism	Attack / threat mitigated	Real-world example
Confidentiality	Only authorized people can read data	Encryption	Eavesdropping, interception	HTTPS protecting login details
Integrity	Data is not altered undetected	Hash functions, MAC, digital signature	Modification, replay	Checksum verifying a downloaded file
Availability	Systems and data accessible when needed	Redundancy, backup, firewall	DoS / DDoS	CDN absorbing a DDoS attack

Correction: traffic analysis is a passive attack (it only observes, it does not alter anything).

Q2. Substitution Techniques Comparison

Cipher	Key usage	Complexity	Security strength
Caesar	Single shift value (1-25)	Trivial	Very weak: brute force in 25 tries
Monoalphabetic	Permutation of 26 letters (26! keys)	Simple	Weak: broken by frequency analysis
Vigenere	Keyword repeated over text (polyalphabetic)	Moderate	Medium: hides frequencies, but Kasiski / Friedman tests find key length
Hill	n x n invertible key matrix mod 26	Matrix multiplication	Strongest of the four: encrypts letter blocks together, hides single-letter frequency; breakable by known-plaintext attack since it is linear

Most secure: Hill cipher, because it works on blocks of letters, so single-letter frequency patterns are hidden. Its weakness is that it is linear, so a known-plaintext attack can recover the key matrix.

Q3. Substitution vs Transposition

Aspect	Substitution	Transposition
What changes	Symbols are replaced; positions stay	Positions are rearranged; symbols stay
Data structure view	Lookup table / array mapping	Permutation / matrix of indexes
Effect on security	Hides letter identity; frequency hidden only in polyalphabetic forms	Letter frequency stays the same as plaintext
Typical attack	Frequency analysis	Anagramming / pattern guessing

- **Similarities:** both are classical, symmetric, key-based and reversible; combining them gives stronger product ciphers.
- **More vulnerable:** simple substitution, because frequency analysis breaks it quickly. Transposition is harder to attack on long messages but still leaks letter frequencies.

Q4. Steganography Concepts

Steganography = hiding the existence of information. Branches: **Data hiding** (embed secret data; goal: capacity) and **Watermarking** (embed ownership/authenticity; goal: robustness). Both lead to **Secure communication**.

- **(a) Relationships:** steganography hides the existence of information; data hiding embeds a secret message in a cover file; watermarking is a special form of data hiding that embeds ownership information of the cover itself; survivability is the property that lets the hidden data stay detectable after changes.
- **(b) Contribution to secure communication:** data hiding conceals the message, watermarking proves ownership/authenticity, survivability keeps the hidden data intact after compression or editing.

Concept	Real-world application
Data hiding	Concealing a message inside a photo
Watermarking	Copyright protection of stock photos
Survivability	Invisible watermark still detectable after an image is compressed and re-uploaded

Q5. Number Theory in Cryptography

Concept map chain: Modular inverse ← Extended Euclid (computes it) ← Euler's phi (counts coprimes) → Euler's theorem → Fermat's little theorem (prime case) → RSA.

Concept	Statement	Role in encryption / decryption
Modular inverse	a^{-1} mod n exists only if gcd(a, n) = 1	Gives the private key d as inverse of e
Extended Euclid	Finds x, y with ax + ny = 1	Computes the modular inverse efficiently
Euler's phi, phi(n)	Count of numbers below n coprime to n; phi(pq) = (p-1)(q-1)	Defines the modulus for exponents in RSA
Euler's theorem	$a^{phi(n)}$ = 1 (mod n) when gcd(a, n) = 1	Why decryption returns the original message
Fermat's little theorem	a^{p-1} = 1 (mod p) for prime p	Special case of Euler's theorem; basis of primality tests

(a) Math connection: Euler's theorem generalises Fermat's; it uses phi(n), and the inverse of a mod n equals $a^{phi(n)-1}$ mod n, which Extended Euclid computes quickly.
(b) RSA role: n = pq, phi(n) = (p-1)(q-1). Choose e coprime to phi(n); find d with Extended Euclid so that ed = 1 (mod phi(n)). Encrypt $c = m^e$ mod n; decrypt $m = c^d$ mod n.

(c) Most critical: Euler's theorem with phi(n). It guarantees that decryption recovers the message, and keeping phi(n) secret (which requires factoring n) is what protects the private key.